COURSE DESCRIPTION			
Program	**Architecture, Automobile Engineering, Civil Engineering, Civil Engineering & Planning, Civil & Rural Engineering, Civil & Environmental Engineering, Fire Technology and Safety, Food Processing Technology, Hotel Management and Catering Technology, Mechanical Engineering (Automobile Engg.), Mechatronics, Mechanical Engineering, Polymer Technology, Printing Technology, Renewable Energy, Civil Engineering (Construction Technology), Tool & Die Engineering, Wood and Paper Technology**		
Course Title	**Applied Physics for Mechanical, Structural and Industrial Applications**		
Course Code	**2002A**	**Semester**	**I / II**
Type of Course	**Practicum**	**Contact Hours**	**90**
Teaching Scheme (L: T:P:R)	**3:0:3:0**	**Credits**	**4.5**
CIE Marks	**40**	**ESE Marks**	**60**

Introduction

Applied Physics is a core subject in the diploma engineering curriculum that provides a foundation in basic physical principles and their practical applications in engineering and technology. The course covers topics such as mechanics, properties of matter, optics, electricity, magnetism, and modern physics, with emphasis on conceptual understanding and numerical problem solving. It helps students develop analytical and problem-solving skills essential for laboratory work, industrial applications, and further technical studies.

Course Objectives

1	To provide students with a broad understanding of physical principles of the universe to help them develop critical thinking and quantitative reasoning skills
2	To develop scientific temper which will help the diploma engineers to apply the basic concepts of physics and to solve broad-based engineering problems
3	To supplement the factual knowledge gained in physics by first hand manipulation of apparatus.
4	To provide necessary confidence in handling equipment and thus learn various skills in measurement.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge in Physics			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply principles of mechanics (Newton's laws, work, energy, and power) to analyze motion-related problems and real-life engineering applications.	Apply
CO2	Explain and apply concepts of properties of matter, fluid dynamics, and thermodynamics (elasticity, viscosity, Bernoulli's principle, and heat transfer) to solve numerical problems and interpret practical phenomena.	Apply
CO3	Analyze rotational motion and wave phenomena (SHM, wave characteristics, and related relations) to solve engineering problems and interpret physical systems.	Apply
CO4	Apply principles of optics and electromagnetism (lens systems, electricity, magnetism, and electromagnetic induction) to solve numerical problems and explain working of devices.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	-	-	2	2	2	-	-	-
CO2	3	2	2	-	-	2	2	2	-	-	-
CO3	3	2	2	-	-	2	2	2	-	-	-
CO4	3	2	2	-	-	2	2	2	-	-	-
Total	12	8	8	-	-	8	8	8	-	-	-
Correlation Level	3	2	2	-	-	2	2	2	-	-	-
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
3	0	3	0	6	90	4.5	20	60	80	20	0	20	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Mechanics	Units, Fundamental and Derived units, SI system, Scalars and Vectors, Addition of vectors- Triangle law and Parallelogram law (Statement), Resolution of vectors. Displacement, velocity and acceleration, Equations of motion (no derivation), Newton's first law Inertia, Momentum, Force. Newton's second law, derivation of F=ma, Newton's third law, statement of Law of Conservation Momentum, Application-Recoil of gun. Numerical problems related to force and momentum. Definition of Work, Work done in moving an object on horizontal plane. Energy, Kinetic energy, Potential energy with examples, Law of conservation of energy. Illustration of freely falling body. Power- Definition and various units of power	10	0
Module I	Laboratory experiments	Vernier Calipers: To determine the volume of a spherical / cylindrical /rectangular body by measuring its dimensions using vernier calipers. Screw gauge: To determine diameter of a wire / solid ball / thickness of glass plate using screw gauge. Spherometer: To determine radius of curvature of a spherical surface using a spherometer Parallelogram law of forces: To verify parallelogram law of forces and find the mass of the given body.	0	3 hour per experiment
Module II	Properties of Matter and Thermodynamics	Elasticity- definition of Stress and strain, Hooke's law, modulus of elasticity, Young's modulus, Rigidity modulus and Bulk modulus, Numerical problems related to Young's modulus Fluid dynamics-Streamline flow, turbulent flow, Equation of Continuity (No derivation), Bernoulli's Theorem (No derivation), Atomizer, Viscosity, Viscous force, coefficient of viscosity, Stoke's Law. Heat, Temperature, Different scales of temperature and relation connecting various scales (Numerical problems related to conversion of temperature into different scales), Mode of heat transfer-Conduction, convection and radiation (basic concepts only).	10	0
Module II	Laboratory experiments	Mercury thermometer: To measure room temperature and temperature of a hot bath using mercury thermometer and convert it into different scales. Hooke's law: To verify Hooke's law and determine force constant of a spring using Hooke's law apparatus. Stoke's law: To find the coefficient of viscosity of a given liquid by measuring the terminal velocity of a spherical body. U-tube apparatus: To determine the relative density of liquid using U-tube apparatus.	0	3 hour per experiment
Module III	Rotational and Wave Motion	Circular motion, angular displacement, angular velocity, angular acceleration, Relation between linear velocity and angular velocity, Relation between linear acceleration and angular acceleration. Centripetal acceleration and Centripetal force- definition and equation only. Rigid body, Moment of inertia, radius of gyration for rigid body, Theorems of parallel and perpendicular axes (derivation not needed), Moment of inertia of rod, disc, ring and sphere (hollow and solid)- (Formulae only) Definition of angular momentum and torque, Conservation of angular momentum (quantitative ideas only). Periodic motion, simple harmonic motion (SHM), definition and examples, time period and frequency. Wave motion -Transverse and longitudinal waves with examples, characteristics of a wave – velocity, time period, frequency, wavelength, amplitude and phase. Relation connecting frequency, wave velocity and wavelength and its numerical problems.	11	0

Module III	Laboratory experiments	Simple pendulum: To determine acceleration due to gravity at a place by measuring the time period of a simple pendulum Moment bar: To determine the mass of the given body using moment bar. Spring constant: To determine spring constant of the given spring by oscillation method. Resonance column: To determine the velocity of sound in air at room temperature using resonance column apparatus Flywheel: To find the moment of inertia of a flywheel	0	3 hour per experiment
Module IV	Optics & Electromagnetism	Reflection, Laws of reflection, Refraction, Law of Refraction, Snell’s law, Refractive index, convex lens and concave lens, Focal length, sign convention, Image formation by convex lens, Distance formula connecting u, v and f for lenses, Power of lens, Relation between focal length and power, Numerical problems based on distance formula and power of convex lens. Electric charge, Unit of charge, Coulomb’s law, Electric field, Electric potential, Electric current, Unit, Direct and alternating current, Ohm’s law, Resistance. Basic ideas of Magnetic Field and Magnetic Flux, Magnetic force, Lorentz forces. Concept of electromagnetic induction, Faraday’s laws.	10	0
Module IV	Laboratory experiments	Snell’s law: To verify Snell’s law using a glass slab/glass prism. Convex lens: To determine focal length and power of a convex lens by u-v method. Ohm’s law: To verify Ohm’s law by plotting a graph between current and potential difference.	0	3 hour per experiment

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Units	Classify physical quantities as fundamental and derived and state their SI units	L	CO1	Understand	1
Units	Explain the SI system of units and its significance	L	CO1	Understand	1
Scalars and Vectors	Differentiate scalar and vector quantities with examples	L	CO1	Understand	1
Scalars and Vectors	Explain triangle and parallelogram laws to resolve vectors	L	CO1	Understand	1
Laws of Motion	Explain Newton’s laws of motion with real-life examples	L	CO1	Understand	1
Laws of Motion	Define momentum and state Newton’s second law of motion	L	CO1	Understand	1
Laws of Motion	State law of conservation of momentum	L	CO1	Understand	1
Laws of Motion	Apply conservation of momentum to analyze recoil of a gun	L	CO1	Apply	1
Work, Energy and Power	Define work, energy and power with units	L	CO1	Remember	1
Work, Energy and Power	Explain kinetic and potential energy with examples, law of conservation of energy, and motion of a freely falling body.	L	CO1	Apply	1
Vernier Calipers	To determine the volume of a spherical / cylindrical /rectangular body by measuring its dimensions using vernier calipers.	P	CO1	Apply	3 hours per experiment
Screw gauge	To determine diameter of a wire / solid ball / thickness of glass plate using screw gauge.	P	CO1	Apply	3 hours per experiment
Spherometer	To determine radius of curvature of a spherical surface using a spherometer	P	CO1	Apply	3 hours per experiment
Parallelogram law of forces	To verify parallelogram law of forces and find the mass of the given body.	P	CO1	Apply	3 hours per experiment
Module II					
Elasticity	Define stress and strain and explain Hooke’s law	L	CO2	Understand	1
Elasticity	Classify the various moduli of Elasticity	L	CO2	Understand	1
Elasticity	Solve numerical problems based on Young’s modulus.	L	CO2	Apply	1
Fluid Dynamics	Differentiate streamline and turbulent flow	L	CO2	Understand	1

Fluid Dynamics	Explain equation of continuity and Bernoulli's principle	L	CO2	Understand	1
Fluid Dynamics	Apply Bernoulli’s theorem to explain atomizer	L	CO2	Understand	1
Fluid Dynamics	Explain viscosity and Stoke’s law	L	CO2	Understand	1
Heat and Temperature	Differentiate heat and temperature and explain different temperature scales	L	CO2	Understand	1
Heat and Temperature	Convert temperatures between various scales.	L	CO2	Apply	1
Heat transfer	Explain conduction, convection and radiation as modes of heat transfer.	L	CO2	Understand	1
Mercury thermometer	To measure room temperature and temperature of a hot bath using mercury thermometer and convert it into different scales.	P	CO2	Apply	3 hours per experiment
Hooke’s law	Determine spring constant experimentally	P	CO2	Apply	3 hours per experiment
Stoke’s law	To determine the coefficient of viscosity of a given liquid by measuring the terminal velocity of a spherical body.	P	CO2	Apply	3 hours per experiment
U-tube apparatus	To determine the relative density of liquid	P	CO2	Apply	3 hours per experiment
Module III					
Circular Motion	Explain circular motion and define angular displacement, angular velocity and angular acceleration	L	CO3	Understand	1
Circular Motion	Derive relations between linear and angular velocity and between linear and angular acceleration.	L	CO3	Understand	1
Circular Motion	Define centripetal acceleration and centripetal force.	L	CO3	Understand	1
Rotational motion	Explain rigid body, moment of inertia and radius of gyration.	L	CO3	Understand	1
Rotational motion	State and Explain parallel and perpendicular axis theorems.	L	CO3	Understand	1
Rotational motion	Explain moment of inertia of rod, disc, ring and sphere (solid and hollow).	L	CO3	Understand	1
Rotational motion	Define angular momentum and torque, Discuss conservation of angular momentum	L	CO3	Understand	1
Simple Harmonic motion	Explain periodic motion and simple harmonic motion with examples, time period and frequency.	L	CO3	Understand	1
Wave Motion	Differentiate transverse and longitudinal waves and explain wave characteristics	L	CO3	Understand	1
Wave Motion	Derive the relation connecting frequency, wave velocity and wavelength	L	CO3	Apply	1
Wave Motion	Solve its numerical problems.	L	CO3	Apply	1
Simple pendulum	To determine acceleration due to gravity at a place by measuring the time period of a simple pendulum	P	CO3	Apply	3 hours per experiment
Moment bar	To determine the mass of the given body using the moment bar.	P	CO3	Apply	3 hours per experiment
Spring constant	To determine the given spring by oscillation method.	P	CO3	Apply	3 hours per experiment
Resonance column	To determine the velocity of sound in air at room temperature using resonance column apparatus	P	CO3	Apply	3 hours per experiment
Flywheel	To find the moment of inertia of a flywheel	P	CO3	Apply	3 hours per experiment
Module IV					
Reflection	Explain laws of reflection	L	CO4	Understand	1
Refraction	Explain Snell’s law and refractive index	L	CO4	Understand	1

Lenses	Differentiate convex and concave lenses and define focal length	L	CO4	Understand	1
Lenses	Explain the image formation in convex lens using the sign convention	L	CO4	Understand	1
Lenses	Describe lens formula and power of lens	L	CO4	Understand	1
Lenses	Numerical problems based on distance formula and power of convex lens.	L	CO4	Apply	1
Electricity	Explain electric charge, Coulomb’s law, electric field and potential	L	CO4	Understand	1
Electricity	Differentiate Direct and alternating current and state Ohm’s law and resistance.	L	CO4	Understand	1
Magnetism and Electromagnetic Induction	Explain basic ideas of magnetic field, magnetic flux,	L	CO4	Understand	1
Magnetism and Electromagnetic Induction	State Lorentz force and Faraday’s laws of electromagnetic induction.	L	CO4	Understand	1
Snell’s law	To verify Snell’s law using a glass slab/glass prism.	P	CO4	Apply	3 hours per experiment
Convex lens	To determine focal length and power of a convex lens by u-v method	P	CO4	Apply	3 hours per experiment
Ohm’s law	To verify Ohm’s law by plotting a graph between current and potential difference.	P	CO4	Apply	3 hours per experiment

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
	Module I	
1	List a set of quantities and classify them as fundamental and derived quantities. Give their SI units. With the help of online resources and performing numerical problems refine the understanding of vector addition. With the help of online resources and performing numerical problems refine the understanding on resolution of vectors.	6
2	Solve numerical problems on equations of motion. Give an activity to demonstrate Newton’s first law of motion. Watch a video on inertia and give examples from daily life.	6
3	Give three examples of Newton’s II & III laws from daily life. List electrical appliances at home with their power ratings. Calculate energy consumed by the appliance in 1 hour.	6
4	Calculate the work done during daily activities such as climbing a staircase, taking water from the well etc. Relate the energy intake by food and the energy requirement for various daily activities	6
	Module II	
5	Hooke’s Law at Home (Concept Application) using rubber band and coins (load)- Learn extension $\propto$ load (within elastic limit). Identify and explain 5 real-life applications of elasticity. Numerical Practice (Self-Assessment): Solve 5 numerical problems related with elasticity (Young’s modulus / stress / strain). Concept Map / Mind Map: Create a colourful concept map including: Elasticity, Stress, Strain, Hooke’s law, Elastic moduli (Y, K, η)	6
6	Identify and list 5 real-life examples for each streamline flow and turbulent flow. Water Flow from Bottle: verify the equation of continuity qualitatively by making small and large holes on it. Concept writing: Write the equation of continuity and explain with help of a diagram. Blowing Between Two Paper Strips and find the relation between pressure and velocity. Identify and explain 5 real-life applications of Bernoulli theorem.	6
7	Flow of Different Liquids- compare viscosity of different liquids such as water, cooking oil, honey / glycerine, on a transparent glass plate. Drop Time Experiment: Observe the effect of viscosity on falling liquid drops. Coin in Liquid Experiment: To observe viscous resistance by measuring time taken in different liquids. Temperature and Viscosity: study the effect of temperature on viscosity.	6
8	Ice–Water Mixture - (Concept of temperature during phase change) : heat can be absorbed without temperature change. Heating Different Materials – (Heat capacity concept): compare rise in temperature for equal amounts of different substances keeping in sunlight for the same time. Numerical Practice (Self-Assessment): Solve 5 numerical problems related with conversion of temperature into various scales. Identify and list 5 real-life examples for each conduction, convection and radiation.	6
	Module III	

9	Identify real life example of circular motion and prepare a short note based on it. Watch youtube simulations on simple harmonic motion and note amplitude and frequency Prepare note on periodic motion in engines, vibrations in machines etc.	6
10	Create a comparison chart of moment of inertia different rigid bodies. Watch videos based on angular momentum and attempt a quiz based on this.	6
11	Solve 5 numerical problem relating velocity wavelength and frequency. Prepare a quiz based on Wave Motion.	6
12	Observe and list wave motions in daily life. Use PhET simulations on waves on a string change frequency amplitude and observe the effects on Wave speed and shape.	6
Module IV		
13	Observe the practical examples of refraction. Using online resources understand the image formation in the human eye. Identify and list out devices which uses convex and concave lens.	6
14	Solve 10 numerical problems based on distance formula and power of lens.	6
15	A Short analytical study on – Why AC is preferred over DC in power transmission. Chose any 5 household electrical devices and find –type of current (AC/DC), operating voltage and current, calculate its resistance using Ohm's law	6
16	Application of electromagnetic induction in generators and transformers	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Text Book of Physics for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi	NCERT	
2	Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi	Dr. H H Lal	Tata McGraw Hill, Delhi

REFERENCE			
Sl. No.	Title	Author	Publication
1	Fundamentals of Physics, Halliday/Resnick/Walker, Wiley India Pvt. Ltd	Halliday,Resnickaand Walker	Wiley India Pvt. Ltd

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://phydemo.app	All
2	https://phet.colorado.edu/en	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Vernier Calipers	0.01cm LC	10
2	Screw gauge	0.01mm LC	10
3	Spherometer	0.01mm LC	10
4	Parallelogram law of forces	60X50	5
5	Mercury thermometer	120 degree centi	5
6	Heat bath	1 Litre	2
7	Hooke's law apparatus with weights		2
8	Stop Watch		20
9	U-tube apparatus		5
10	Stoke's law apparatus		5
11	Convex lens with stand, screen and light box wire gauze		5

12	Equilateral Prism/glass slab with drawing board		5
13	Voltmeter, ammeter, rheostat and battery eliminator		8 set
14	Standard resistance	1 and 2 ohm	4 set
15	Semiconductor diode, bread board and micro ammeter	1N4001	5
16	Carbon resistor	various ohms	20
17	Photoelectric cell apparatus		2
18	Simple pendulum apparatus		3 set
19	Resonance column		4
20	Flywheel apparatus with slotted weights		2
21	Slotted weights	50gms	8
22	Tuning Fork	4 different frequencies	4 set

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology – Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	
Exam mark		10	50	40	40	50	50	60	
Converted to			10	10	10	10	10		
Marks	5	5	10	10		10		60	
Total	40							60	
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10

Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

Note:
Experiments shall be conducted such that all the COs are attained.
A minimum of 8 to 10 experiments shall be conducted.